Talktile: Accessible Educational Tool to Augment Independent Learning for Students with BVI

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Abstract. Blindness and visual impairment (BVI) are significant barriers to independent learning. While there have been advancements in assistive technologies, a critical need remains for tailored tools that empower individuals with blindness to access and navigate educational materials independently. This gap was particularly amplified during the COVID-19 lockdown when students primarily relied on self-learning with limited interactions with their instructors. This work addresses the issues and proposes solutions for effective, independent, affordable and equitable learning opportunities for students with BVI through technology. We present a multimodal interactive system that makes widely available printed textbooks accessible by superimposing digital layers of audio annotations, which are triggered based on the position of the user's finger on a page. This tool aims to fill the gap created by the lack of affordable, multimodal and accessible educational resources for independent learning by students with BVI, particularly for remote and self-learning scenarios.

Keywords: Accessible Education · Visual Impairment · Inclusive Design · Voice User Interface · Audio-AR · Machine Learning

1 Introduction

Blindness or Visual Impairment (BVI) is a "low incidence" and "high needs" disability, triggering unique challenges pertaining to inclusion [4]. It calls for skilled instructors and specific measures around training and teaching the affected. Today, 18 crore children are taught by almost 57 lakh teachers in more than 12 lakh primary and upper primary schools nationwide [12]. Inclusive education bridges the opportunity gaps between students with and without physical disabilities.

Our primary stakeholders for this study are high school students with blindness and visual impairment (BVI). School teachers are our secondary stakeholders. We found that most institutions were not well equipped with resources to provide quality high school education. Additionally, most accessible educational material is available either as Braille books or audiobooks. Both formats limit the communication of accompanying graphical information, seriously limiting information in graphic-intensive subjects like geography and biology.

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Following the double-diamond model [17], we defined our problem statement, reviewed the literature, and conducted on-site focus group interviews for localised need identification. We integrated the learnings to design and develop the first iteration (proof-of-concept) of Talktile, which we evaluated using blindfolded roleplay testing methodology. All our cumulative learnings were then used to iteratively build the second and latest version of the Talktile system, which we tested with children with BVI on-site.

Our work presents our overall research, design and development journey. We propose a novel approach to make print books and tactile maps more accessible by introducing layers of informational audio tags triggered using gesture and voice-based navigation. This approach adds a multimodal dimension to independent learning from printed books and tactile graphics. Additionally, the contributions of this work include: 1) Describing the critical usability and accessibility issues in currently existing solutions by gathering insights from literature reviews and need identification studies; 2) Describing the design and technology framework of the Talktile system, its evolution over iterations, and presenting the qualitative and quantitative user evaluation insights.

Considering the challenges faced by the BVI population in India, there is a motivation to contribute to improving their quality of life. The project aims to provide an affordable, portable, and easily accessible way to assist students with BVI, aligning with the economic conditions and limited accessibility to study material in the country. This paper aims to strike consequential discussions around creating novel and affordable interactive systems that aid accessible education.

2 Background

We conducted scoping literature reviews on the relevant research, design and technological developments to understand the range of challenges faced by students with BVI. According to a World Health Organisation (WHO) survey, India accounts for approximately 20% of the world's visually impaired population, a significant number considering its vast population. Despite the alarming figures, it is crucial to understand the challenges faced by visually impaired individuals in the context of India's economic conditions and limited accessibility to medical assistance [13].

Students with BVI encounter difficulties in accessing educational materials, particularly in tool subjects like Braille. The apparent attempts to teach Braille and other tools have limitations, with auditory senses having limited value in providing concrete clues about objects [18]. The lack of material for second language acquisition challenges visually impaired learners. However, there is potential for proficiency in acquiring a second language, especially for those who may not read Braille.

Despite challenges, there is a consensus that visually impaired students do not face serious problems with listening skills. Listening is a primary learning method; these students rely on auditory data to process information effectively. Developing speaking skills is essential for visually impaired students to request

and transfer information. These skills enable effective communication and participation in academic activities.

Visually impaired students often face limitations in their knowledge regarding various concepts and areas, impacting their academic performance. Addressing this challenge is vital for their active participation in assignments and group discussions.

Emphasising the need for inclusion in the Indian education system, there is a call to enrol children with visual and hearing impairments into mainstream schools. The National Association for the Blind (NAB) in India is crucial in advocating for inclusive education, ensuring that every visually impaired child has access to formal education [20].

Despite efforts like Braille and audio translations, accessing print media remains challenging. Tactile diagrams considered the closest to vision in terms of semantic and cognitive processing, have been explored to communicate graphical information [16]. Tactile diagrams or tactile graphics (TGs) are significant in making graphical information accessible to blind individuals.

An increasingly popular alternative to braille labels is the use of audio labels, which enable audio-haptic exploration of the object, whereby the user explores the surface naturally with the fingers and queries locations of interest; note that blind people typically prefer to explore objects with multiple fingers or hands instead of just one [14], which is compatible with the use of audio labels.

3 Need Validation Study

Through our literature review, we found that the previous works and existing solutions were over budget, sparsely available and required specialised hardware, making them inaccessible in a scenario that calls for a highly accessible, budget-friendly, easy-to-use technological intervention- a gap we aim to fill with Talktile. Our primary objective with the need validation study was understanding the stakeholders' needs, pain points and wants.

3.1 Methodology

We conducted on-site focus group interviews at three blind institutions/ associations to comprehensively understand the challenges faced by students with BVI. We performed the pilot testing of the focus group interviews at NAB India Centre for Blind Women and Disability Studies (NABICBWD), New Delhi [6], which gave us a general overview. We conducted the following interviews at All India Confederation of the Blinds (AICB), New Delhi [1] and Bhartiya Blind School (BBS), New Delhi [2]. We interviewed 52 students and 3 teachers after getting their informed verbal and written consent. We used a mixture of English and Hindi for the interviews, and the transcripts were later translated verbatim into English for analysis.

Each session, consisting of semi-structured interviews, lasted 70–90 min in focus group sets of 12, 14, and 26 students. The questionnaire comprised general

queries about the students' backgrounds, school life, classes and subjects and detailed queries about the challenges faced while transitioning between inclusive schools and blind schools. Participants expressed anxiety and sadness while narrating personal experiences and challenges.

Understanding the instructor's point of view was crucial as they are the primary facilitators in their students' learning. The semi-structured one-on-one interviews lasted for around 30–45 min. We gathered insights into their problems while instructing and mentoring students with BVI.

3.2 Demographics

NABICBWD is a women-only association. We interviewed 14 students [14F] aged 15–20 years. Most students completed their schooling during the COVID-19 pandemic and were pursuing their higher education at the institution.

At AICB, we interviewed 12 students [5F, 7M] aged 16–20 years. All students were enrolled in high school (grades 9–12) at the time of the study. Most students were comparatively older than the sighted students enrolled in the same grades, owing to difficulties taking classes due to their disability. We also interviewed one school teacher.

At BBS, we interviewed 26 students aged 16–19 years and two school teachers. This institution was one of the select few schools active in-person during the pandemic (Fig. 1).



Fig. 1. Need identification studies at A) NABICBWD, B) AICB, and C) BBS. Participants' faces are covered to conceal their identities.

3.3 Findings

Our engagement with students with BVI unveiled poignant narratives. The challenges faced by students extended beyond the classroom, encompassing limitations within the educational system and adapting to online education. The impact of curriculum design on practical courses, the struggle with inaccessible websites and materials, and the unique issues encountered in inclusive educational settings came to the forefront. Students expressed a desire for more tailored support, detailing the strain of competing with sighted peers and grappling

with inaccessible educational content. These revelations underscore the urgent need for inclusive practices, acknowledging the diversity in learning capacities among visually impaired students.

The insights from interviewing teachers provided a broader canvas, offering perspectives on how they navigate the challenges faced by students with VI and guide them. Four distinct sub-themes emerged, highlighting the limitations of existing resources, issues surrounding accessibility, and the diverse methods employed by teachers to impart knowledge effectively. Teachers illuminated the struggles faced by students with limited resources, emphasising the gap between available tools and the capacity to utilise them. Accessibility challenges were underscored, especially in the realm of graphics and visual content, posing hurdles in comprehension and practical engagement. Teachers used innovative methods, such as visualising concepts, teaching special symbols, and emphasising fundamental building blocks, which emerged as crucial strategies to bridge the learning gap.

4 Proof of Concept Prototype

4.1 Design and Working

Need identification insights set the guiding design principles for developing the proof of concept on Unity utilising Vuforia SDK [5,10]. The first iteration of Talktile, as seen in (Fig. 3), featured a custom-designed, easy-to-assemble MDF stand with adjustable height and a phone mount. A mobile phone device running our AR application sat atop and above the appropriate educational material. The Vuforia SDK uses a computer vision-enabled image-detection algorithm and compares the input image with a database uploaded on its cloud network. Using Unity, we overlaid virtual buttons on top of image targets on a printed science textbook and a political map of India as our sample educational materials. Finger-based interactions triggered the appropriate virtual buttons and played back audio tags relevant to the target beneath.

For independent learning using the first iteration of Talktile, users place the learning material on the base of the assembled stand. Then, they choose the chosen subject by clicking its virtual button on the phone screen. The rest of the interactions are natural, as the user moves their finger across the relevant texts or graphics on the placed material.

4.2 User Evaluation of Prototype and Findings

Our primary objective with the initial evaluation was to gain insights into the system's efficacy in engaging students and usability problems. We conducted blindfolded roleplay tests with 12 participants [8M, 4F] to cover the temporary and situational blindness spectrum and comply with the limitations of the COVID-19 lockdown. During the evaluation, the blindfolded participants adjusted to the new environment with an ice-breaking session, following which



Fig. 2. Design and setup for Talktile PoC (first iteration).

they studied the provided educational resources using our prototype. Each session lasted 10 min, after which we asked them qualitative questions. Insights mainly were in favour of the prototype, with around 70% of participants completing the tasks. Around 50% found the UI buttons small, and 10% found the solution a hassle due to the steep learning curve. Other notable limitations of this iteration were related to the touchscreen GUI nature of the interface (Fig. 2).



Fig. 3. Blindfolded roleplay testing of Talktile PoC (first iteration).

5 Proposed Interactive System - "Talktile"

5.1 Design and Working

Encompassing all previous learnings, we developed the design and framework of the latest and the proposed iteration of the Talktile interface. It features a laser-cut MDF stand with more height and phone position adjustments than the outgoing version. The audio tags augment the supplied NCERT textbook and thermoformed tactile maps (Fig. 4B) we procured from RLF [9]. The student places their smartphone atop the low-cost stand. The smartphone is the hub for processing the gestural and voice-based user inputs and educational material through the camera feed, running our application, and playing back the audio outputs.

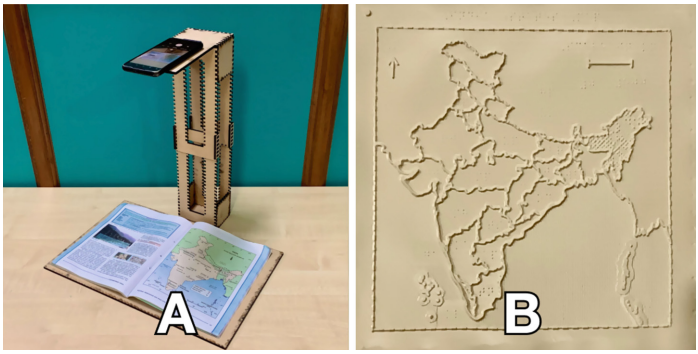


Fig. 4. A) Design and setup for Talktile, the proposed interactive system. B) Thermoformed tactile political map of India.



Fig. 5. Sample screenshots during user testing with Talktile. The coloured boxes seen here are the annotation regions to help identify audio interaction to be played within a region.

Our application integrates technologies, including TensorFlow [11] for model training, TFLite for model inference on smartphones, and Android Studio for application logic development. Its capabilities include page identification within a book and precise hand position tracking on a page or a tactile map. These

features are instrumental in enhancing the accessibility of user interactions with printed books and tactile graphics.

The application utilises an easy-to-use speech-based interaction method. Users have the ability to speak out voice commands, and in response, the application provides auditory outputs. Examples are provided in (Table 1). The voice-user interface (VUI) provides seamless and organic interactions between the user and the machine. It addresses the constraints and steep learning curve of a gesture-based navigation system. The Voice User Interface (VUI) utilises the wake word detection system provided by PicoVoice Porcupine [8]. This enables the application to efficiently and precisely identify user commands and initiate the corresponding actions.

The machine learning model can be used on a smartphone without being connected to the internet.

Table 1. Sample user interactions on the second iteration of Talktile.

Interaction Type	Command	Response/Action
Open a new page	Automatic (no command)	"Chapter 5. Page 39, 40."
Bring hand over a section	Automatic (no command)	"Paragraph - Government steps to protect flora and fauna..."
Reading out loud	"Read this"	"To protect the flora and fauna of the country, the government has taken many steps. 1. Eighteen biosphere reserves have..."
Summarise text	"Explain"	"Describe the government's steps to protect flora and fauna, including biosphere reserves and conservation projects."
Describe image/figure	"Explain"	"Figure 5 shows a Montane Forest with conical trees..."

5.2 Vision Model Development

Our neural network used an EfficientNetV2 B0 backbone [19] for picture information extraction. It was trained to do three tasks: page identification, hand presence identification, and hand position localization.

The hand-position dataset was created using the FreiHAND dataset, a hand-segmentation task dataset [21]. Multiple hand images were masked and overlaid onto page images. We trained our vision model on this data, and sample screenshots of its working are shown in (Fig. 5).

5.3 User Evaluation and Findings

Our primary objectives were to investigate how students with BVI interacted and what they felt about the iterative version of Talktile, and its efficacy in creating

an independent learning experience. We performed the study with ten participants [10M], aged 12–16 years [Mean: 14.3, SD: 1.25], at the National Association for the Blind, New Delhi [7]. Each session was conducted after taking informed verbal and written consent. The study was divided into four major sections: Pre-Test Discussion, Prototype Testing after briefing them about the system, Technology Acceptance Model (TAM) Testing [15], and Post-test Discussion. The open-ended semi-structured discussions covered the methods participants previously used for independent learning scenarios, their experience interacting with the Talktile system, and how they felt about it. The prototype testing measured the user's ability and time taken to perform three tasks: **Onboarding**, where the user had to open the book to a prompted chapter; **Textbook Reading**, where they had to use the Talktile system to read 2 text paragraphs and 1 figure from the chapter; **Map Reading**, where they had to switch from the textbook to the tactile map, use the system to read map data and answer 2 questions we asked them about it (Fig. 6).

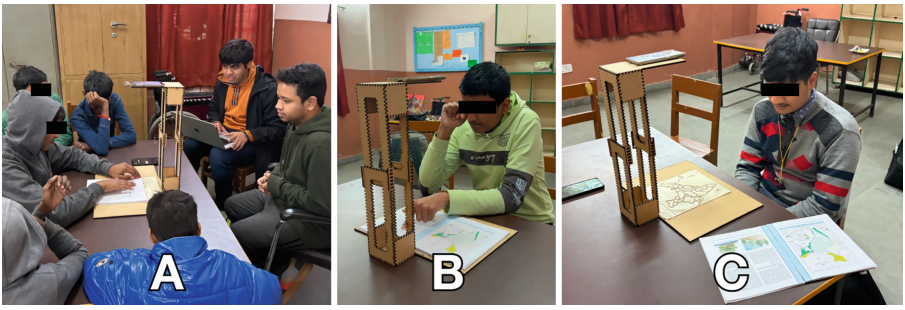


Fig. 6. User evaluation of Talktile, our latest iteration. Participants' faces are covered to conceal their identities.

We formulated a Technology Acceptance Model (TAM) questionnaire consisting of 6 prompts about Perceived Usefulness (PU) and another 6 prompts about Perceived Ease-of-Use (PEU), as shown in (Table 2).

Our analysis of the TAM survey data for the Talktile system (Fig. 7) uncovered insightful findings about user perceptions. The data, collected on a Likert scale of 1 to 7, with 1 being the most favourable, indicated mean ratings for all variables ranging from 1.8 to 2.6. This range reflects a generally positive reaction to the system. Notably, the most positive reception was observed for PU1, which relates to the system's ability to enable quick task accomplishment (Mean = 1.8, SD = 0.8), and PU4, which measures the enhancement of effectiveness in independent learning (Mean = 1.8, SD = 1.2). The relatively lower standard deviation for PU1 suggests a more consistent agreement among users regarding this aspect of the system. Conversely, areas for potential improvement

Table 2. Descriptive Statistics for Technology Acceptance Model (TAM)

TAM Variable	TAM Prompt	Mean	SD
PU1	Using the Talktile system in my independent learning would enable me to accomplish tasks more quickly.	1.8	.8
PU2	Using the Talktile system would improve my independent learning performance.	2.3	1.6
PU3	Using the Talktile system in my independent learning would increase my productivity.	2.4	1.5
PU4	Using the Talktile system would enhance my effectiveness in independent learning.	1.8	1.2
PU5	Using the Talktile system would make it easier to do my independent learning.	2.1	1.4
PU6	I would find the Talktile system useful in my independent learning.	2.5	1.6
PEU1	Learning to operate the Talktile system would be easy for me.	2.1	1.7
PEU2	I would find it easy to get the Talktile system to do what I want it to do.	2.6	2.1
PEU3	My interaction with the Talktile system would be clear and understandable.	2.3	1.3
PEU4	I would find the Talktile system would be clear and understandable.	2.2	1.2
PEU5	It would be easy for me to become skilful at using the Talktile system.	2.0	1.9
PEU6	I would find the Talktile system easy to use.	1.9	1.4

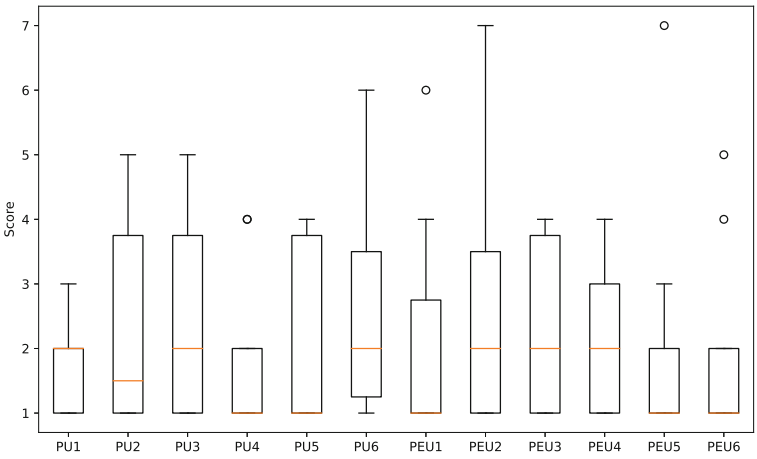


Fig. 7. Box Plot of TAM Survey

were identified, notably in PEU2 (ease of getting the system to do what one wants, Mean = 2.6, SD = 2.1), where the higher standard deviations indicate a wider variability in user experiences and perceptions.

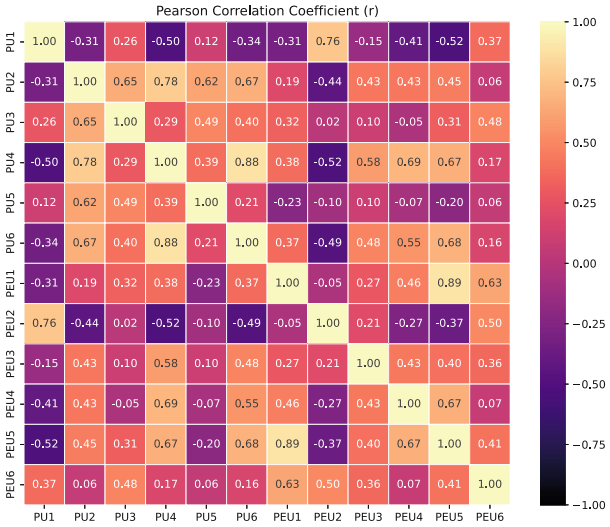


Fig. 8. Pearson Correlation Coefficient (r) matrix of the TAM survey results.

Correlation analysis from data in (Fig. 8) revealed strong positive relationships between several key variables, such as between PEU5 (ease of becoming skillful) and PEU1 (ease of learning to operate) with an r value of 0.886. This suggests a robust link between the ease of learning the system and becoming skilled in its use. However, certain negative correlations emerged, notably between PEU5 (ease of becoming skilful) and PU1 (quick task completion) with an r value of −0.523, and between PEU2 (ease of getting the system to do what one wants) and PU4 (effectiveness) with an r value of −0.516. These negative correlations suggest a possible trade-off between the perceived effectiveness of the system and its ease of use or speed in task completion.

These negative correlations underscore the importance of enhanced user training. Effective training can help users better understand and navigate the system. This approach can mitigate the perception that effectiveness requires a steeper learning curve. By focusing on practical, hands-on training that emphasizes best practices for using the system, users can become more adept at navigating the system effectively.

The overall correlation between the averages of Perceived Usefulness (PUs) and Perceived Ease of Use (PEUs) was found to be moderate, at 0.317. This moderate correlation indicates that while there is some association between the system's perceived usefulness and its ease of use, the relationship is not as strong as might be anticipated. This finding has important implications for system

development and user experience design. It suggests that enhancing the system's usefulness might not necessarily lead to proportional improvements in its ease of use, and vice versa. Therefore, further efforts to make the system more useful should not compromise its usability, and improvements in ease of use should not overlook the system's utility.

In conclusion, while the TAM survey data for the Talktile system indicates a generally positive reception, there are specific areas identified for improvement. The system is well-regarded for its effectiveness in enhancing independent learning and facilitating quick task completion. However, the identified trade-offs between effectiveness and ease of use or speed highlight the need for enhanced user training. Such training could lead to increased user satisfaction and a more comprehensive acceptance of the system.

5.4 Notable Challenges

Participants were surprised by the responsiveness of the system, but they noted that it would be beneficial only in the absence of braille books or ebooks (depending on their preference), and that it would not replace their primary method of studying. They agreed that it would be especially beneficial when braille texts become scarce at the start of each academic year. Some demonstrated a greater interest in the application's underlying technology and suggested that the Text-to-Speech narrator be equipped with additional media controls (pause, stop, replay, forward, and rewind) and control over playback speed. Participants tended to utilise both hands when engaging with the Talktile interface, suggesting a requirement for a more intuitive and user-friendly design that assumes two-handed usage. They expressed concern about not knowing if they had covered all the information on a page, raising questions about the effectiveness of the tactile learning experience. Instead of smoothly dragging their finger across the page, some students tended to jump across, potentially leading to a disjointed understanding of the content. Furthermore, they expressed difficulties determining the subsequent paragraph's location upon finishing one, indicating a desire for enhanced navigation indicators. Several participants encountered challenges in comprehending the exact boundaries of geographical elements on tactile maps, highlighting a potential limitation in the Map Reading component of the current iteration of Talktile.

6 Discussion

In this research, we have presented a novel multimodal intervention aimed at improving the independent learning process for students with BVI. We drew insights from the need identification study provided with the breadth of problems and challenges faced by students with BVI and their teachers. We explored the challenges faced by students with BVI in pursuing higher education, particularly in STEM fields. Applying these learnings, we built the first iteration of the Talktile interface as a proof of concept. To address the situationally blind

user spectrum and late-blindness to some degree while adhering to COVID-19 lockdown constraints, we conducted blindfold roleplay testing. By integrating all prior knowledge, we iteratively developed the second and proposed version of Talktile and assessed its performance at a facility for visually impaired individuals.

In general, the participants had developed a habitual approach to education in which they had no difficulty self-studying so long as they had access to pertinent material in their preferred formats, such as braille books or laptops with ePub files. The assistance of a sighted individual was only required if they had any uncertainties concerning the material's content. They had all previously utilised a tactile board that their instructor employed to instruct on shapes and diagrams; thus, they had all been acquainted with tactile graphics.

Insights primarily indicated that Talktile addressed the gaps that were present in learning from printed non-Braille textbooks. Students also noted how the interface allowed them to gather more knowledge from tactile graphics, as they could attribute informational data to their respective geographical locations on the tactile map.

However, Talktile still has unexplored potential. The final testing of the interface reveals various future development tangents, including a mature Voice-User Interface capable of effectively communicating with students and providing them with additional information on demand. We learned about navigability challenges students had when studying with our system, such as a lack of knowledge of the layout and boundaries of printed content and using both hands as they normally do when engaging with their surroundings. Enhanced tutorials pertaining to the system's operation will effectively address these challenges.

7 Future Work and Conclusion

Existing large language models (LLMs) such as ChatGPT [3] or a custom model can be fine-tuned to allow students to explore the out-of-book knowledge and ideas, which can take the independent learning experience to a new level. The potential for automating the image labelling and explanation generation process is considerable. Future enhancements can further integrate document extraction models and image caption generators to streamline the content preparation process. These developments would improve the efficiency and expand the application's capabilities and, ultimately, the learning experience.

The proposed multimodal Talktile interface effectively enhances the independent learning experience for students with BVI. We discussed what we learned through the different research and development stages and highlighted user study results. The evolution of the interface is prominent through the two presented iterations. Future explorations into this idea can add many more better-featured functionalities, ultimately making opportunities around education more equitable and accessible. We hope this work sparks consequential ideas and discussions around the specific interaction methods and accessible education in general. We envision a future where all students are afforded equal access to educational opportunities.

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